5G NR Pilot-Free Joint Neural Demodulator & LDPC Decoder

Master Project Defense & Meeting Preparation Guide (Everything You Must Know)

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+16.7%	2 dB	0% Errors	0% Pilots
Spectral Efficiency Gain	Power / Coding Gain	at 6 dB Eb/N0	100% Data Grid

■ YOUR 30-SECOND ELEVATOR PITCH TO PROFESSOR SHERVIN VAKILI

"Professor, in this project we implemented an end-to-end 5G NR physical layer transceiver that completely eliminates pilot symbol overhead (14.3% in standard 5G). We designed a lightweight 2D ResCNN demapper (~90k parameters) that ingests the 2D time-frequency resource grid and directly outputs soft LLRs to the 5G LDPC decoder. To enable online self-supervised training without labels, we designed a decision-directed Delta-loop where the LDPC decoder acts as an oracle to supervise the network via an Experience Replay Buffer. Our simulations prove a +16.7% spectral efficiency gain and a 2 dB coding gain, achieving zero post-FEC errors at 6 dB where standard 5G still suffers from a 59.4% frame error rate."

1. PROFESSOR'S 4 TASKS: WHAT, WHY & HOW WE IMPLEMENTED THEM

Exact Mapping of Assigned Tasks to Code Implementation and Results

Task # & Requirement	Why We Need It (The Logic)	How We Implemented It in Code	Key Verification / Result
Task 1: 5G NR OFDM + LDPC Pipeline (Pilot-Free)	To send maximum data bits per slot without wasting 14% of bandwidth on pilot tones.	OFDMFECPipeline in ofdm_fec_chain.py: 14x64 grid (896 REs), rate-1/2 LDPC (k=1792, n=3584), 16-QAM mapping, pilot_pattern=None.	+16.7% Spectral Efficiency (2.00 vs 1.714 b/s/Hz).
Task 2: Classical 3GPP Reference Baseline	To have an exact fair benchmark against what standard 5G base stations do today.	ClassicalPilotBaseline in classical_baseline.py: Kronecker DMRS at symbols 2 & 11 (128 pilots), LS channel estimation, LMMSE equalization, APP demapping.	Benchmark reference: reaches 0 error at 8 dB (needs 2 dB more SNR).
Task 3: 2D ResCNN Neural Demapper	To replace separate channel estimation and equalization with a single spatial neural demapper.	ResCNNDemapper in neural_demapper.py: Ingests [B, 2, 14, 64], 48 hidden channels, 3 ResBlocks with GELU and GroupNorm (G=1), outputs 3584 calibrated LLRs.	Raw demapper BER is 4-7% lower across all SNRs before FEC.
Task 4: Self-Supervised Delta Trainer & Replay Buffer	To let the AI adapt online in real-time without needing true ground-truth labels or pilot tones.	JointDemodFECTrainer & ExperienceReplayBuffer in trainer.py and replay_buffer.py: LDPC BP decoder corrects errors -> pseudo-codeword -> BCE Delta loss -> 256-frame buffer.	Stable online adaptation without catastrophic forgetting.

2. END-TO-END PHYSICAL CHAIN: FROM BITS TO AIR TO RECOVERY

Step-by-Step Flow with Tensor Shapes and Physical Explanations

Processing Stage	Data / Tensor Shape	Simple Explanation (What Happens Here)
1. Information Bits (u)	[B, 1792] binary {0, 1}	Your original payload message (e.g. digital image pixels or file bytes).
2. 5G NR LDPC Encoder	[B, 3584] codeword c	Adds 1792 mathematical parity check bits (code rate R = 0.5) to enable error correction.
3. 16-QAM Modulator	[B, 896] complex symbols	Groups every 4 bits into 1 complex constellation point on a 4x4 grid (Amplitude I + Phase Q).
4. Resource Grid (IFFT/CP)	[B, 14, 64] time-frequency	Places 896 symbols into 14 OFDM symbols x 64 subcarriers. Zero pilots (100% data). Adds 16-sample CP.

5. Wireless Channel (Air)	[B, 14, 64] complex Y	Signal is distorted by multipath fading H (echoes) and corrupted by background static noise W (AWGN).
6. 2D ResCNN Demapper	[B, 3584] soft LLRs	Splits Y into Real/Imag [B, 2, 14, 64], extracts spatial correlations, outputs calibrated soft bit confidences.
7. 5G LDPC BP Decoder	[B, 1792] recovered u_hat	Runs 20 iterations of Belief Propagation message-passing to correct bit errors and output clean payload.

3. KEY RESULTS: BIT ERROR RATE & SPECTRAL EFFICIENCY

Verified Monte Carlo Simulation Data Across Signal-to-Noise Ratios

Eb/N0 (dB)	Classical Baseline Post-FEC BER	Classical Baseline BLER	Proposed Neural Post-FEC BER	Proposed Neural BLER	Key Finding / Takeaway
0 dB	0.3283 (32.8%)	1.0000 (100%)	0.2918 (29.2%)	1.0000 (100%)	High noise regime
2 dB	0.2717 (27.2%)	1.0000 (100%)	0.2483 (24.8%)	1.0000 (100%)	Early decoding region
4 dB	0.2044 (20.4%)	1.0000 (100%)	0.1132 (11.3%)	0.9219 (92.2%)	Neural cuts error in half
6 dB	0.0175 (1.75%)	0.5938 (59.4%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	**2 dB Coding Gain: 0 errors vs 59.4% frame failures!**
8 dB	0.0000 (0.0%)	0.0000 (0.0%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	Both error-free; Neural carries +16.7% more data
10 dB	0.0000 (0.0%)	0.0000 (0.0%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	Both error-free; Neural carries +16.7% more data

Operating SNR	Transmission System	Bit Error Rate (BER)	Measured PSNR (dB)	Visual Image Recovery Quality
5 dB	Raw Channel (No FEC)	15.00%	12.10 dB	Heavy salt-and-pepper noise
5 dB	Classical 3GPP Baseline	11.96%	16.56 dB	Blurry with block artifacts
5 dB	**Proposed Neural Demapper**	**0.46%**	**26.17 dB**	**Clear and recognizable (+9.6 dB PSNR gain)**
6 dB	Classical 3GPP Baseline	2.04%	25.98 dB	Minor visible pixel error clusters
6 dB	**Proposed Neural Demapper**	**0.058%**	**38.49 dB**	**Crystal clear reconstruction (+12.5 dB PSNR gain)**
8 dB	**Proposed Neural Demapper**	**0.000%**	**inf dB (Perfect)**	**Bit-for-bit perfect recovery with +16.7% faster speed**

4. PROFESSOR Q&A; DEFENSE: HOW TO ANSWER HIS QUESTIONS

Direct, Professional Answers to Expected Theoretical & Practical Questions

Q1: Why is a 2D CNN better than classical Least Squares (LS) channel estimation?

A1: Classical LS estimation only looks at the 128 pilot subcarriers and performs noisy interpolation, leaving substantial estimation error. Our 2D ResCNN looks at the ENTIRE 14x64 resource grid simultaneously (896 observations). Because adjacent subcarriers and symbols share coherence in time and frequency, the CNN extracts 2D spatial correlations, performing joint channel denoising and demapping without pilots.

Q2: How can the neural network train online without true ground-truth labels?

A2: We use a decision-directed self-supervised Delta loop. The 5G LDPC Belief Propagation decoder acts as an oracle. When the CNN outputs soft LLRs, the LDPC decoder corrects random bit errors using its parity equations. We re-encode those clean bits into a parity-consistent codeword (H*c^T = 0), which serves as a high-fidelity pseudo-ground-truth label to update the CNN weights via backpropagation.

Q3: What prevents the neural network from catastrophic forgetting during online updates?

A3: Two mechanisms: (1) A Reliability Filter with threshold tau = 30% that discards frames from deep fades where the LDPC decoder failed, and (2) A circular Experience Replay Buffer of capacity 256 frames. We sample random mini-batches of 32 stored frames for AdamW updates, breaking temporal correlation and preserving past channel knowledge.

Q4: What is the computational complexity of the 2D ResCNN?

A4: It is very lightweight: only ~90,000 parameters across 3 residual blocks with 48 hidden channels. It uses Group Normalization (G=1) and GELU activations, enabling low-latency execution suitable for 5G baseband hardware.

Q5: Where does the +16.7% spectral efficiency gain come from mathematically?

A5: In standard 5G, 128 out of 896 resource elements are dedicated to Kronecker pilots (14.3% overhead), giving 768 data REs (k=1536 bits, eta = 1.714 b/s/Hz). In our pilot-free design, all 896 REs carry data (k=1792 bits, eta = 2.000 b/s/Hz). The gain is (2.000 - 1.714) / 1.714 = +16.71% extra payload at zero additional power or bandwidth.